

Programming Using C++ Level 1

Conditional Else IF statements

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Conditional Else IF statements

Definition

- ◆ An else if statement allows a program to check multiple conditions in sequence.
- ◆ The program checks each condition from top to bottom and executes the first condition that is true.

Explanation

◆ This program receives a grade and assigns a letter based on its value.

■ 90-100 → A

■ 80-89 → B

■ 70-79 → C

■ 60-69 → D

■ 50-59 → E

■ Less than 50 → F

◆ If none of the valid conditions is satisfied, the else statement prints Invalid grade.

Steps in main()

1. Get the grade

◆ Read the user's grade and store it in the grade variable.

2. Check for A

◆ If the grade is between 90 and 100, print A.

3. Check for B

◆ Otherwise, if the grade is between 80 and 89, print B.

4. Check for C

◆ Otherwise, if the grade is between 70 and 79, print C.

5. Check for D

◆ Otherwise, if the grade is between 60 and 69, print D.

6. Check for E

◆ Otherwise, if the grade is between 50 and 59, print E.

7. Check for F

◆ Otherwise, if the grade is less than 50, print F.

8. Handle invalid input

◆ If none of the conditions is true, print Invalid grade.

💡 Example

- ◆ If the user enters:
 grade = 85
- ◆ The program checks the conditions from top to bottom.
- ◆ 85 is not between 90 and 100.
- ◆ 85 is between 80 and 89, so the program prints:
 B

➡ Result

- ◆ Example results:
 - Grade: 95 → A
 - Grade: 85 → B
 - Grade: 75 → C
 - Grade: 65 → D
 - Grade: 55 → E
 - Grade: 40 → F
 - Grade: 105 → Invalid grade

Note

- ◆ Conditions in an else if chain are checked from top to bottom.
- ◆ Once a condition is true, the remaining else if and else statements are skipped.
- ◆ The && operator is used to require both limits of a range to be true.

⊘ Warning

- ◆ The order of conditions matters when using else if.
- ◆ Make sure the conditions cover the intended range and handle invalid values when necessary.



Comparison



if

- ✓ Checks the first condition.
- ✓ Executes if the condition is true.
- ✓ Starts the decision-making chain.



else if

- ✓ Checks another condition if the previous condition was false.
- ✓ Allows multiple possible conditions.
- ✓ Only the first true condition is executed.



else

- ✓ Executes when all previous conditions are false.
- ✓ Does not have a condition.
- ✓ Provides a final alternative.

✦✦ Summary

- ✓ else if allows a program to check multiple conditions in sequence.
- ✓ Conditions are evaluated from top to bottom, and the first true condition is executed.
- ✓ else can handle cases where none of the previous conditions is true.

Thank you 😊